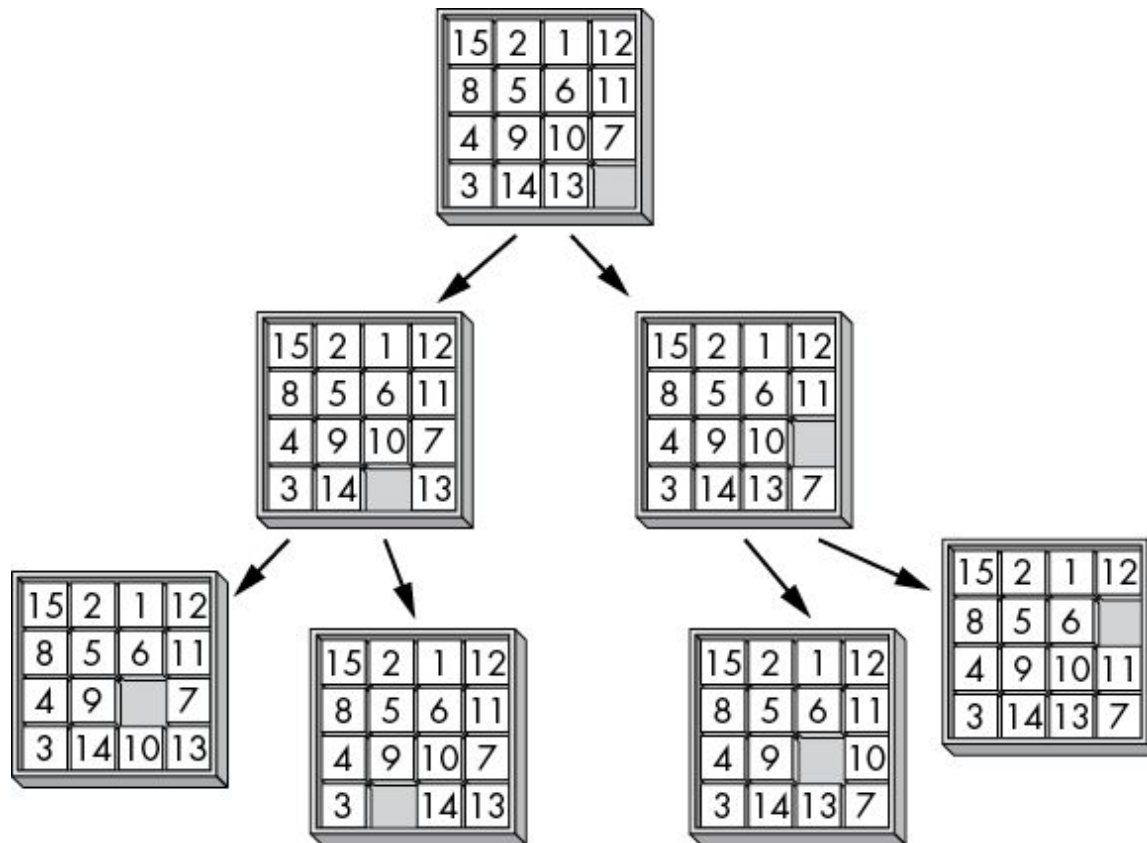




*Figure 12-2: The task of solving a 15-puzzle can be represented as a graph with tile states as nodes and slides as edges.*

Clever algorithms are available for solving 15-puzzles, but we could also just recursively explore the entire tree graph until we find a path from the root node to the solution node. This puzzle's tree can be searched with a depth-first search (DFS) algorithm. However, unlike a well-connected maze, the 15-puzzle's tree graph is not a DAG. Rather, the graph's nodes are *undirected*, because you can traverse both directions of an edge by undoing the previous slide you made.

[Figure 12-3](#) shows an example of the undirected edges between two nodes. Because it is possible to go back and forth between these two nodes forever, our 15-puzzle algorithm could encounter a stack overflow before it finds a solution.